

SCI-JINC

Scientific Rationale and Contract

Contract v0.1 / document r0.3

2026-08-29

Status. Implementation-blind Stage B author draft. The r0.2 architecture is owner-approved as the basis for this revision, and Disposition A with the paired center tie is owner-approved. Final Stage B acceptance, implementation-blind consistency review, and freeze remain pending. No implementation conformity, representation fidelity, validation, achieved performance, numerical readiness, production readiness, or production claim is made.

Input	Exact admitted PTC transformed occurrences and positive JINC-permitted PTC coefficients; exact same-processed-sample AST coordinates; one authorized array-associated JINC parameter set; one exact positive-integer phase-bin count; one compatible target WCS; finite positive processed-sample frequency.
Output	Zero through three independent observation bundles, at most one for each requested and admitted stable array. Each complete bundle contains N_p , C_p , Q_p , derived m_p with local support/validity, and $T_p^{(\kappa^2)}$.
Core equation	$m_p = (\sum_i I_{ip} \omega_i \kappa_{ip} z_i) / (\sum_i I_{ip} \omega_i \kappa_{ip})$.
Scientific sources	Approved SCI-JINC Scope Brief and sanitized boundary packet; recovered implementation-independent signed-estimator core under its supersession cover; Schloerb LMT OTF/JINC method excerpt under its TolTEC-exclusion cover.
Availability	The generic scientific contract is defined. No TolTEC numerical parameter set and no registered, selected, realized JINC-permitted PTC coefficient family/payload is admitted or supplied by this source packet. No exact owner-approved numerical-adequacy profile or matching certificate is admitted or supplied either. Consequently no numerical route or numerical adequacy is claimed by this draft.

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1 Program adherence and prior-work recovery

This package follows the Citlali Scientific Contract Library Program, its pilot process controls, and the owner-approved downstream sequence recorded by the sanitized Scope Brief. It begins after the frozen SCI-MAP scientific authority without altering it. The author received only the exact approved SCI-JINC packet, the exact external objects admitted through packet covers, the exact targeted r0.2 repair directive, the exact final targeted r0.3 owner-review directive, and the owner’s explicit disposition-A approval. No implementation, configuration, test, audit, repair, validation, reduction, status, production, or manager-side record supplied normative scientific content. Non-packet records were restricted to process and exact-byte closure. The admitted packet, the two targeted directives, and the exact owner disposition remain the complete scientific authority for this repair.

Prior work is reused rather than repeated. The recovered signed estimator with distinct numerator, signed normalization, and quadratic accumulator is adopted. The generic OTF/JINC analytic family is cited through the admitted Schloerb method excerpt [3]. The earlier circular radial-cutoff and pixel-area-average branches are superseded by the approved fully populated square cache and point-phase operator. Response and covariance mathematics is preserved conditionally for future science, but no such base-v0.1 product is created. Historical implementation behavior, parameter recommendations, audits, and achieved results remain excluded.

The genuinely new Stage B work is therefore documentary and contractual: one shared scientific authority is expressed as this rationale and the companion Engineering Conformance Specification. The shared modules define notation, definitions, equations, assumptions, requirements, and falsifiable predictions. No estimator re-derivation is performed merely to make the work appear new.

The exact source closure used by both views is reported in Appendix A. The manifest remains the exhaustive authority for all sixteen admitted objects; the appendix records byte and process closure without enlarging the scientific source set.

2 Physical model and scientific intent

An on-the-fly observation samples the sky irregularly. The telescope and upstream processing determine which spatial information is present in those samples; gridding then estimates values on a regular destination WCS. The generic JINC motivation is spatial-frequency control: the peak-normalized Fourier partner of a circular aperture passband is a signed JINC function [3]. A practical analytic family multiplies one JINC factor by a generalized-exponential envelope and a second JINC factor. The second factor places a controlled zero while the envelope damps distant sidelobes.

The scientific object is not a nearest-neighbor average and not an ordinary positive interpolation kernel. Positive lobes, analytic zeros, and negative lobes are all part of the operator. The sign carries physical mathematical meaning: a negative lobe subtracts from both the signal numerator and the normalization, while it enters the quadratic statistic and coefficient-squared time through its square. Removing that sign would change the estimator.

The generic analytic identity is completely specified by Equations (2) and (3). The angular separation is normalized by an explicit array-associated scale s_a . The dimensionless tuple $(a_a, b_a, c_a, (r_{\max})_a)$ controls the two JINC arguments and envelope. This defines what each parameter means; it does not select TolTEC values. The Schloerb excerpt supplies a generic LMT method, not TolTEC array tuning, validation, or performance authority.

3 Scientific boundary and ownership

SCI-JINC begins with one exact PTC occurrence and the AST continuous coordinate for the same processed sample realization. PTC owns the transformed quantity, retention, cleaning realization, immutable lineage, and the one versioned registry of positive analysis/gridding coefficient families. Every family declares permission for named consumers; permission for SCI-MAP does not imply permission for SCI-JINC. The accepted request selects one explicitly permitted family, or one exact owner-authorized versioned mode policy selects it. No ambient or implementation default is authorized. Requested, effective, observation-resolved, and realized identities remain distinct.

AST owns the coordinate, frame, WCS, parent-sample association, validity, uncertainty, and producer causes. SCI-JINC does not reconstruct pointing or infer a coordinate from row order, time tolerance, detector order, shape, or numerical equality. It consumes the AST coordinate only after the exact same-processed-sample relation is established.

SCI-JINC owns JINC upstream-occurrence admission (Registry identity `SCI-JINC:jinc_map_contribution01`), local offset and radial geometry, center rounding, point phase, square membership, signed coefficient, normalization, cancellation conditioning, fixed accumulators, local map support, grouping, destination identity, and atomic publication. It applies upstream facts without changing their meanings. A producer cause has an effect only when an exact restriction names it; generic flags do not acquire a universal downstream veto.

This division follows the producer-transformer-consumer rule. ALIGN/AST, RTC, CAL, PTC, and VAL retain their own meanings. Later noise, filtering, beam, source, observing-mode, and recurrence packages retain theirs. SCI-JINC does not create those sciences at the gridding boundary.

4 Analytic kernel, point phase, and finite geometry

The target is a finite rectangular FITS pixel grid $\mathcal{D} = \{1, \dots, N_x\} \times \{1, \dots, N_y\}$ with integer pixel centres. The exact AST coordinate for occurrence i is $\mathbf{u}_i = (x_i, y_i)$ in those same one-based continuous pixel coordinates. Centre resolution is componentwise:

$$c_{i\mu} = \lfloor u_{i\mu} + \tfrac{1}{2} \rfloor, \quad \phi_{i\mu} = u_{i\mu} - c_{i\mu} \in [-\tfrac{1}{2}, \tfrac{1}{2}), \quad \mu \in \{1, 2\}.$$

An exact half-pixel tie therefore selects the larger integer and has residual phase $-1/2$. The scientific owner retains this positive-axis directional tie as an intended part of the discrete operator. The resolved WCS axis order and signs are normative: reflecting or reindexing an axis creates a different target-WCS identity rather than an alternative interpretation of this tie rule.

Let $n_{\text{sub}} = \text{subpixel_n}$ be any exact positive integer selected by the accepted request or by one exact owner-authorized versioned mode policy. Each axis is partitioned into equal left-closed, right-open bins. The logical bin and its midpoint representative are

$$q_{i\mu} = \lfloor n_{\text{sub}}(\phi_{i\mu} + \tfrac{1}{2}) \rfloor, \quad \hat{\phi}_{i\mu} = -\tfrac{1}{2} + \frac{q_{i\mu} + \tfrac{1}{2}}{n_{\text{sub}}}.$$

Thus $q_{i\mu} \in \{0, \dots, n_{\text{sub}} - 1\}$, and a phase exactly on an interior bin boundary selects the upper bin. No parity restriction is imposed. In particular, an even n_{sub} maps $\phi = 0$ to the representative $+1/(2n_{\text{sub}})$; that result is disclosed and normative, not repaired by a hidden special case. The scientific owner approved disposition A: the upper-bin convention and every positive integer n_{sub} are retained, including this one-sided exact-boundary phase offset. The mathematical cache key is the pair of bin indices and

the two integer offsets. Physical cache layout is an engineering choice only when it preserves that exact logical mapping.

The target WCS binds its celestial frame and topology, projection, reference values, one-based reference pixel, dimensions, units, ordered axes, signs, rotation, handedness, version, and a constant nonsingular 2×2 pixel matrix $\mathbf{M}$. Base v0.1 accepts only square, equal-scale, orthogonal pixels:

$$\mathbf{M}^\top \mathbf{M} = \Delta^2 \mathbf{I}, \quad \Delta > 0.$$

Equivalently, $\mathbf{M} = \Delta \mathbf{R}$ with $\mathbf{R} \in O(2)$; either handedness is permitted and recorded. The contractual angular separation is the Euclidean norm in this exact tangent plane. A skewed or anisotropic target WCS is typed incompatible; a scalar pixel size may not conceal it. This restriction narrows which AST-owned WCS realizations JINC accepts and does not alter AST authority.

For array a ,

$$h_a = \left\lceil \frac{s_a(r_{\max})_a}{\Delta} \right\rceil, \quad \mathbf{d} \in \{-h_a, \dots, +h_a\}^2.$$

The destination pixel centre is $\mathbf{p} = \mathbf{c}_i + \mathbf{d}$. The exact normative discrete point-phase radius is

$$\hat{r}_{i\mathbf{d}} = \left\| \mathbf{M}(\mathbf{d} - \hat{\phi}_i) \right\|_2, \quad \hat{r}'_{i\mathbf{d}} = \hat{r}_{i\mathbf{d}}/s_a,$$

and Equation (3) is evaluated at that radius. The unquantized continuous phase is not a hidden reference oracle; the midpoint-quantized operator above is the scientific operator. Its exact-real-arithmetic evaluation is the oracle for a later numerical certificate.

The disposition is directly observable at an exact integer pixel center. For even n_{sub} , zero phase on both axes has representative $(1/(2n_{\text{sub}}), 1/(2n_{\text{sub}}))$. At the nominal central offset $\mathbf{d} = \mathbf{0}$, square-pixel orthogonality gives

$$\hat{r}_{i\mathbf{0}} = \frac{\Delta}{\sqrt{2}n_{\text{sub}}}, \quad \kappa_{i,\mathbf{c}_i} = \kappa_a \left(\frac{\Delta}{\sqrt{2}n_{\text{sub}}s_a}; a_a, b_a, c_a, (r_{\max})_a \right).$$

This is the intended discrete scientific operator. The nominal central cache element is not replaced by the zero-radius limit $\kappa_a(0) = 1$.

The exact constant in the second JINC factor is $j_{1,1} = \min\{x > 0 : J_1(x) = 0\}$. A rounded decimal is at most nonnormative guidance for an evaluator. This definition makes the second factor exactly zero at $r' = (r_{\max})_a$ while leaving analytic continuation beyond that radius intact.

The square window is fully populated. The half-width formula is a square truncation rule, not a circular support predicate: outer rows and corners can lie beyond $s_a(r_{\max})_a$ and are still evaluated. Geometry first requires $\mathbf{c}_i \in \mathcal{D}$. If it fails, the occurrence contributes nowhere. If it passes, a square member is retained exactly when $\mathbf{p} \in \mathcal{D}$. There is no radial cutoff, wrap, reflection, completion, edge correction, or edge renormalization. Direct construction on a smaller map therefore need not equal larger-map construction followed by a crop.

5 Signed estimator and cancellation

The upstream ω_i is positive. SCI-JINC supplies the signed spatial factor κ_{ip} and forms $w_{ip} = \omega_i \kappa_{ip}$. For every destination pixel it accumulates distinct N_p , C_p , and Q_p using the same occurrence membership and coefficient realization. The normalized estimate is $m_p = N_p/C_p$, not N_p/Q_p . This choice gives unit response to a constant input wherever the signed normalization is accepted.

Signed cancellation is the central validity issue. A finite negative C_p is scientifically admissible. Exact $C_p = 0$ is unavailable, not zero sky. The dimensionless statistic $\rho_p = |C_p|/D_p$ describes the degree of cancellation without introducing a unit-bearing threshold. It is an indicator, not a universal cutoff.

Algebraic support and numerical-realization adequacy are distinct. Algebraic support requires finite inputs and exact discrete coefficients, $Q_p > 0$, and $C_p \neq 0$. Numerical adequacy additionally requires one exact owner-approved, versioned profile and one matching certificate bound to the realization. A profile must identify its scientific observable, absolute or relative error form, reference magnitude or norm, complete parameter/source/ pixel domain, numerator and denominator treatment, near-zero behavior, simultaneous or worst-case aggregation, exact inequality, certification method, and lifecycle/provenance. The certificate must identify the exact discrete oracle, realization, source bytes, WCS, coefficient and parameter states, and prove that the profile's inequality passes on its declared domain.

For a certificate that bounds denominator error by $|\hat{C}_p - C_p| \leq \epsilon_{C,p}$, numerical support requires the exact separation condition $|\hat{C}_p| > \epsilon_{C,p}$. If it does not hold, zero remains inside the certified denominator interval and the normalized value is unavailable. Any additional normalized-map error inequality comes from the exact profile; this contract does not invent one. A profile without a matching certificate, or a certificate without its exact profile, cannot establish adequacy.

A passing $\mathcal{P}_{\text{NA}}/\mathcal{E}_p$ pair has a deliberately narrow claim boundary: it establishes only finite-precision fidelity to the exact selected discrete SCI-JINC oracle over the declared domain. It does not establish scientific adequacy of s_a , a_a , b_a , c_a , $(r_{\text{max}})_a$, n_{sub} , or square truncation; source-response or beam fidelity; photometric or astrometric performance; optimality; observational validation; response/covariance fidelity; readiness; or production authority. Qualifying a TolTEC parameter set remains a separate scientific and observational program with its own objective, evidence, authority, and lifecycle.

No exact owner-approved numerical-adequacy profile or matching certificate is admitted or supplied by this source packet. Numerical-adequacy state, and in particular numerical near-cancellation support, is therefore typed unavailable in this draft. No qualitative instrument-fidelity phrase, universal ρ_p cutoff, unit-bearing floor, contributor-count/machine-epsilon rule, summation algorithm, reduction order, or bitwise requirement substitutes for that absent authority.

6 Products, grouping, and lifecycle

Every produced bundle has exactly five required roles. Three are primary accumulator planes: N_p , C_p , and Q_p . The fourth is derived m_p with its local support/validity state. The fifth is $T_p^{(\kappa^2)}$, the `jinc_coefficient_squared_time` coefficient-squared temporal-accounting product. It accounts for the squared analytic coefficient divided by processed-sample frequency. It is not acquired exposure, complete temporal support, precision, validity, or significance.

Table 1: Fixed product meanings and prohibited reinterpretations.

Role	Scientific meaning	It is not
Signal numerator	Signed sum $N_p = \sum_i I_{ip} \omega_i \kappa_{ip} z_i$.	A normalized map, response, or significance plane.
Signed normalization	Signed sum $C_p = \sum_i I_{ip} \omega_i \kappa_{ip}$.	Exposure, positive weight, support, or a quantity that may be forced positive.

Role	Scientific meaning	It is not
Quadratic accumulator	$Q_p = \sum_i I_{ip} \omega_i \kappa_{ip}^2$, required by algebraic local support.	Automatic variance, precision, exposure, or validity.
JINC map	$m_p = N_p / C_p$ on accepted local support, in the admitted signal unit.	Empirical significance, a source fit, or a response companion.
Coefficient-squared time	$T_p^{(\kappa^2)}$ in seconds, accounting for the squared analytic coefficient.	Physical acquired exposure, hits, white-noise-equivalent time, normalized influence, or confidence.

A whole-product failure in any required role suppresses the bundle. A local invalid pixel does not: local support is ordinary map content. The fixed schema therefore needs neither a generic role-availability object nor a placeholder product. It also creates no response, covariance, formal-weight, physical-exposure, diagnostic, or generalized provenance role.

6.1 ODQ-107 scope parity and deferred companions

The targeted r0.2 scope-parity check compared the draft with the exact successor Scope Brief admitted by the manifest, rather than with an earlier or unlisted scope statement. That exact source records the owner-approved ODQ-107 amendment: the base bundle is the closed five-role numerical schema above, while response, covariance, formal weight, generalized companion availability, and generalized provenance products are deferred. The pre-repair draft's narrowed product list was therefore not an author-created departure from its admitted Stage A authority.

The targeted directive's conditional broadening branch does not apply because the exact scientific-owner amendment already exists in the admitted successor Scope Brief and Q002-approved bytes. No new Stage A amendment is needed to retain REQ-038-040. Any future decision to replace ODQ-107 would require a bounded successor Stage A amendment and renewed exact-byte owner approval. This r0.3 draft retains the conditional mathematics and records the static deferred scope honestly in Appendix B. Fixed-state, PTC-procedure, JINC-re-resolved, and whole-chain response remain distinct; covariance and formal-weight identities remain conditional. None is a base-v0.1 product or per-bundle information-state record.

Observation is the scientific grouping boundary. One observation may produce zero through three independent bundles, at most one for each of `a1100`, `a1400`, and `a2000`. A missing, unavailable, or unrequested array creates no placeholder and does not invalidate a different array. Contributions with different array or destination identities never merge.

Same-observation samples or processing chunks may realize one bundle only under one exact observation, array, JINC realization, target WCS, admission/parameter/coefficient state, destination, and lifecycle generation. A chunk, worker, process, container, filename, or vector position is not a scientific product identity. Cross-observation combination is not defined. A future JINC coadd must begin from complete observation bundles under a separate scientific boundary.

6.2 Compact replayable generative record

Reproducibility does not require dense per-contribution provenance or a sixth numerical plane. It does require one compact generative record at the existing plan/bundle provenance granularity. That record binds the exact contract and source manifest; requested, effective, observation-resolved, and realized identities; observation and occurrence-population rule; PTC and AST parents; immutable admission-profile decision; coefficient family, generation, payload and permission; parameter set; target WCS and metric; rounding, tie, phase-bin and midpoint rules; square extent and edge membership; destination; accumulator and cancellation rules; numerical profile and certificate state; required-role completion/failure; lifecycle generation; and the fixed ODQ-107 scope identity under which no response/covariance companion information-state record exists.

Replay reconstructs the exact candidate occurrence population from the bound parents, reapplies the immutable upstream-admission decision, resolves the same centre, phase bin, representative, square, and cropped destination membership, evaluates the exact discrete kernel, and then forms the coupled accumulators in the declared numerical realization. The record must make a changed source, plan, coefficient, parameter, WCS, phase count, profile, or generation distinguishable rather than treating it as the same result. This is compact generative lineage, not an authorization for a new generalized provenance product or a per-pixel contribution archive.

7 Validity, uncertainty, and claim separation

Algebraic local support requires finite admitted inputs, finite N_p , C_p , and Q_p , $Q_p > 0$, and $C_p \neq 0$. Certified numerical-realization adequacy is a separate state and is typed unavailable without the exact profile and certificate. Empirical downstream policy may narrow these states but cannot promote an invalid pixel. Outside square support, an analytic zero, and an outside rounded center are ordinary no-contribution states. Missing or ambiguous coordinate association is different: geometry is unavailable and must not be relabeled as outside support.

The formal variance and covariance equations are retained because they are part of the recovered signed-estimator science. They are conditional on the exact coefficient family and covariance assumptions. With dimensionless ω_i , Q_p/C_p^2 is dimensionless and is not signal variance. Even with inverse-square units, units alone do not prove inverse variance or independence. No formal-weight or covariance plane is therefore authorized for the fixed bundle.

Table 2: Uncertainty budget and present evidentiary state.

Source	Scientific treatment	Present state
Signal and calibration	Retain PTC/CAL quantity, calibration, response, uncertainty, quality, and lineage as producer facts.	Not re-estimated or validated by SCI-JINC.
Astrometry	Retain AST coordinate uncertainty, Jacobian availability, WCS, and causes.	No JINC astrometric performance claim.

Source	Scientific treatment	Present state
PTC coefficient	Retain the selected family's statistic, normalization, support, uncertainty, QC, and covariance meaning.	A realized JINC-permitted family and payload are not admitted or supplied by this source packet.
Kernel parameters	Bind an exact array-associated parameter-set identity.	TolTEC numerical values are not admitted or supplied by this source packet; the route is unavailable.
Finite numerical realization	Bind one exact owner-approved profile and one matching realization certificate satisfying its exact inequality.	No such profile or certificate is admitted or supplied; adequacy is typed unavailable, not tested or achieved.
Response and covariance	Preserve conditional identities for later use.	No base-v0.1 product or information-state role under ODQ-107 and no achieved response/covariance claim.
Observational performance	Requires later authorized numerical and observational validation.	Not performed.

Table 3: Claim layers remain separate.

Layer	What it would establish	Status of this draft
Algebraic contract correctness	Internal consistency of identities, equations, units, limits, and predictions.	Drafted and mechanically reviewable; not yet owner-approved or frozen.
Implementation conformity	A candidate realizes the binding requirements.	Not assessed; no candidate inspected.
Representation fidelity	Stored or transmitted products preserve the scientific identities and meanings.	Not assessed.
Numerical and observational validation	Realized error, response, covariance, photometry, and observation behavior satisfy authorized profiles.	Not performed.
Achieved performance	Measured scientific performance under named conditions.	No claim.
Readiness and production authorization	Operational qualification and explicit owner authorization.	No claim.

8 Limiting cases and interpretive safeguards

The signed estimator has simple limiting cases that expose its meaning without requiring an implementation-specific realization. For a constant field, $z_i = z_0$ factors out of the numerator, so

$N_p = z_0 C_p$ and $m_p = z_0$ wherever the normalization is accepted. For one contributor with nonzero κ_{ip} , the signed coefficient cancels between numerator and normalization and the estimate returns that contributor's signal. These are normalization identities, not achieved photometric-validation claims.

Common positive scaling of every ω_i changes the three primary accumulators together but leaves m_p , ρ_p , and the normalized operator unchanged. This distinguishes coefficient normalization from scientific signal normalization. A signal-unit change has its own transformation: when an exact family truly represents inverse variance, scaling the signal by λ requires scaling the coefficient by λ^{-2} . The map then scales by λ and the conditional formal weight by λ^{-2} .

Table 4: Interpretation of representative limiting cases.

Case	Algebraic result	Scientific safeguard
Constant input	$N_p = z_0 C_p$ and $m_p = z_0$ on accepted support.	Tests the signed N/C normalization, not achieved calibration.
One nonzero contributor	$m_p = z_i$ for either sign of κ_{ip} .	A negative lobe is not an invalid sample.
Analytic zero	The occurrence changes none of N_p , C_p , Q_p , or $T_p^{(\kappa^2)}$.	The zero remains distinguishable from outside support.
Finite negative lobe	N_p and C_p retain sign; Q_p and time receive positive squared terms.	Absolute value or clipping changes the operator.
Finite negative normalization	$m_p = N_p/C_p$ remains algebraically available if every other gate passes.	Denominator sign alone is not a validity cut.
Exact cancellation	$C_p = 0$ makes m_p unavailable.	Never substitute zero sky, infinity, or a valid non-finite value.
Near cancellation	Availability depends on the demonstrated total numerical error under one exact owner-approved profile and matching certificate.	With no admitted profile/certificate, numerical near-cancellation support is typed unavailable; no universal ρ_p threshold is inferred.
Outside rounded center	All $I_{ip} = 0$, even with footprint overlap.	The center-domain gate precedes square support.
In-map cropped edge	Retained pixels use their actual membership and signed N/C normalization.	No wrap, completion, edge correction, or JINC-then-crop requirement.
Common coefficient scale	N_p , C_p , and Q_p scale together; m_p and ρ_p do not.	Coefficient normalization cannot silently change the map.

Coefficient-squared time has a different limiting structure. Its term is zero at an analytic zero and positive for either sign of a nonzero JINC lobe; changing ω_i alone does not change it. Those facts

are exactly why the product cannot be reinterpreted as weighted exposure, precision, or normalized influence.

The edge rules deliberately make finite-map construction part of the observation estimator. An occurrence with an outside rounded center is excluded before footprint evaluation, while an occurrence with an inside center uses a cropped square. Therefore a larger-map JINC construction followed by cropping is not required to equal direct construction on a smaller map. This is a scientific boundary decision, not a numerical defect.

9 Future evidence route without a validation claim

The companion engineering view converts the shared requirements and predictions into an evidence checklist, but no evidence is evaluated during Stage B authorship. A future assessment must first bind the exact coefficient family, TolTEC parameter set, observation, array, WCS, plan, destination, and lifecycle generation. Without those identities, a numerical example cannot stand in for the scientific route.

Algebraic checks should begin with hand-constructed constant, one-contributor, two-contributor, analytic-zero, negative-lobe, and cancellation cases. Geometry checks should then exercise every map edge and corner, half-pixel tie, phase-bin edge, `subpixel_n=1`, square-cache corner, and center immediately inside and outside the finite domain. These constructions establish the operator being tested. They shall use the exact half-tie, half-open-bin, midpoint, WCS-metric, square-extent, and crop rules; there is no alternate tie or bin algorithm within base v0.1.

Numerical adequacy requires the exact profile and certificate described in Section 5. The reference is exact-real evaluation of the normative midpoint-quantized discrete operator. Phase quantization is therefore part of the operator, not realization error relative to an unlisted continuous-phase target. Analytic-function evaluation, cache/index realization, finite arithmetic, accumulation/reduction, and normalized division must be covered exactly as the profile declares. Agreement between two executions is useful evidence but is not itself the scientific criterion: two paths could agree while sharing the same wrong membership or normalization.

Table 5: Future evidence layers and their present limits.

Layer	Representative evidence	What it cannot establish alone
Authority binding	Exact parameter, coefficient, observation, WCS, plan, destination, and lifecycle identities.	Numerical correctness or achieved performance.
Algebra and units	Hand cases, scaling identities, zero limit, signed lobes, and conditional unit classifications.	Implementation conformity without a bound candidate.
Geometry and membership	Tie/bin neighborhoods, square corners, outside-center gate, inside-edge crop, and exact AST association cases.	Observational response or photometry.

Layer	Representative evidence	What it cannot establish alone
Numerical adequacy	Exact owner-approved profile, matching certificate, declared oracle/domain/error functional, and adversarial cancellation cases.	Bitwise reproducibility, observational validity, or readiness.
Bundle/lifecycle	Array permutation, missing-array, destination collision, chunking, atomic failure, and forbidden-role cases.	Representation fidelity without stored-product evidence.
Representation	Exact stored identities, units, WCS, support state, and role semantics.	Observational validation or production authorization.
Observation	Separately authorized response, covariance, photometric, and science-performance evidence.	Readiness or production authority without an explicit decision.

A fail-closed test may demonstrate that missing parameter or coefficient authority prevents a bundle. It may not manufacture the missing authority in order to exercise a numerical path. Likewise, a future response or covariance study must first receive separate product authority; base-v0.1 conditional equations do not turn an observed output into an authorized companion product.

10 Typed unavailability and next scientific dependencies

The contract defines the generic algebra and operator, but three authorities needed for a numerically adequate route are not admitted or supplied by this source packet. First, an exact PTC coefficient family must be registered, selected, realized, and explicitly permitted for SCI-JINC. Second, an exact TolTEC array-associated parameter set must be scientifically authorized. Third, one exact owner-approved numerical-adequacy profile and a matching realization-bound certificate must exist. None of these absences is repaired by memory or software behavior.

No inherited 45 m denominator, inherited shape values, or mode-dependent $r_{\max}$ values are authorized. No value in the generic Schloerb excerpt is a TolTEC default, recommendation, optimum, validation result, or performance claim. Selecting TolTEC parameters is a separate scientific exercise with a named objective and appropriate TolTEC/LMT beam, response, and/or noise evidence.

The targeted repair finds no unresolved Stage A scope discrepancy: the exact ODQ-107 five-role amendment remains controlling. A future proposal to replace it with broader response, covariance, companion-availability, or provenance information-state roles would require a bounded Stage A successor and renewed exact-byte owner approval. Physical exposure, cross-observation coaddition, and broader diagnostics remain deferred until a concrete scientific use separately authorizes them.

A Exact source closure detail

Table 6: Implementation-blind source closure.

Source object	Exact binding	Stage B use
Author-packet manifest	Commit 88dcce8b0f7b1d78053b2 5831b39cf370afd47cc; SHA-256 52a8e843456a8cb033b7593d9b9f67f b83b0ee565c91c141d8e16d46b906140e; all sixteen admitted-object digests byte-equal.	Exhaustive scientific information firewall.
Targeted r0.2 repair directive	SHA-256 c07505861d91459f69e7d0989f11551e 2a14265c916cd5772ea48a86bb186ed2.	Bounded repair instructions and exact operator choices; the admitted ODQ-107 amendment controls the conditional scope branch. The resulting r0.2 architecture is owner-approved as the basis for r0.3.
Final targeted r0.3 owner-review directive	SHA-256 4878e1745e085b4e33d2e71f1190299 d72f2cfd7b2215e36a9e8405a977bd207.	Normative final repair scope and claim boundaries.
Owner phase-lattice disposition	Exact statement “I approve disposition A”, SHA-256 9a70cbc63c0c79a7db70ad9796481fb0 fe3f1f4c2d7524820b54cec68b8b1620.	Retains every positive n_{sub} , the upper-bin even-lattice zero-phase offset, and the paired positive-axis center-tie convention.
Successor Scope Brief	SHA-256 5f2505c2760fc5cb07506249f33f4496 51aca67cccc9c444305b059674f0ddbd.	Scientific scope, including ODQ-107.
Recovered signed core and supersession cover	Core SHA-256 2c1f9ff95f65422a098846f747ed165d 5aedd5bedd854678bfa7faeebba4e24; cover SHA-256 5be650ce4e25f161211955f60696033a1 8adb83d7ff0c9155a776b4e184d601e.	Signed-estimator science under the cover only.

Source object	Exact binding	Stage B use
Schloerb method excerpt and exclusion cover	Excerpt SHA-256 a065843b4b83c21aabb25233c588817e9 98773a5d6a7bd389874eab50c9a88e9; cover SHA-256 9b32095fc7e1773e13e70b4c21d4f402 b0c7376aff0b776d41fa9f5a263b7c4f.	Generic LMT method only; no TolTEC values.
PTC-to-JINC boundary	SCI-PTC_TO_SCI-JINC v0.1/r0.3; SHA-256 5769d413460e931745e0d401ea432b12d 1077c15466247c49caa71b997d4ab1e.	Exact transformed-signal and coefficient boundary.
AST-to-JINC boundary	SCI-AST_TO_SCI-JINC v0.1/r0.2; SHA-256 efffa7059b59c89793fa1d523fb3bb482 35f1ab55f7d55060af1600cbfd470a5.	Same-processed- sample continuous- coordinate boundary.
JINC upstream-occurrence admission	Immutable Registry identity SCI-JINC:jinc_map_contribution@1; packet-object SHA-256 2db95da7e5d1b980df79993907d45ac0 ababc3aa05c189bfb62dcf04ff2c2e8a.	Upstream occurrence consideration only, never pixel support.
VAL Registry binding and successors	Binding SHA-256 ee8f20db5febdb51e39f7157449d6c2d 03a0d17058605dd5531e9ab5ca439e30; SCI-VAL_SOURCE_BINDING_REGISTER successor v0.1/r0.3-map-r0.7. 1-jinc-stage-a-q002-2026-08-28, SHA-256 0e7ca29ee2e9cd02fb1b76cf87cc64fc e6164407a7801f9b9a105ca646317e88; SCI-VAL_PROFILE_REGISTRY successor with the same version, SHA-256 4b9a1ebecfc847c83b59da772afd9b03 1ab1830e8febbb12d1a47f70ce5a1110.	Exact process bindings only; they create no JINC coefficient family, TolTEC parameter set, or numerical-adequacy profile. These JINC-specific snapshot files are exact-source locked; no ambient current Registry is substituted.

B Deferred response and covariance detail

Table 7: Response, covariance, and companion-role disposition.

Family	Science preserved in v0.1	Product/information-state disposition
Fixed-state response	The normalized fixed operator may transform one exact available processed source template exactly once.	Deferred by ODQ-107; no base response request, availability, family, parent, or numerical product role.
PTC procedure response	A future authority may distinguish a full PTC finite difference with the JINC operator held fixed.	Deferred; no hidden inheritance from an upstream or ordinary-MAP response.
JINC re-resolved response	A future authority may re-resolve admission, coefficient, parameter, phase, support, cancellation, or destination state.	Deferred; it is not equivalent to the fixed-state operator.
Whole-chain response	A complete RTC–CAL–PTC–JINC rerun would require separate end-to-end authority.	Unavailable unless that authority and exact rerun are separately supplied.
Covariance	The identity $\mathbf{C}_{\text{JINC}} = \mathbf{A}_{\text{JINC}} \mathbf{C}_{\text{PTC}} \mathbf{A}_{\text{JINC}}^{\top}$ is preserved on an exact available domain.	No base covariance state or product. Missing covariance is never zero or independence.
Conditional diagonal variance/weight	Q_p/C_p^2 and C_p^2/Q_p retain their conditional algebra under a true inverse-variance family and declared independence.	No base product; units or a historical label cannot activate the interpretation.
Companion availability and generalized provenance	Existing producer facts, causes, and bundle/plan identity remain bound.	No generic companion role or generalized product framework under ODQ-107; a successor amendment is needed.

C Canonical notation and units

The symbols in Table 8 are normative in both views of this contract. The generic signal unit is denoted by U . The admitted upstream boundary presently names a nonpolarimetric total-intensity-equivalent quantity in **mJy/beam**; the symbol U keeps the algebra explicit and does not turn a component token into a physical Stokes assertion.

Table 8: Canonical notation.

Symbol	Meaning and domain	Unit
a	Stable TolTEC array identity, exactly one of a1100 , a1400 , or a2000 .	identity
i	Exact admitted PTC occurrence on stable RTC output sample n , with observation, detector occurrence/UID, PTC application generation, segment, array, and immutable parents.	identity
p	Target pixel in one exact JINC destination WCS.	identity
$\mathbf{u}_i = (x_i, y_i)$	Exact finite AST continuous FITS pixel coordinate for occurrence i in the target JINC WCS.	pixel
$\mathbf{c}_i$	Integer FITS pixel-center coordinate selected from $\mathbf{u}_i$ by Equation (4).	pixel
ϕ_i	Exact residual phase $\mathbf{u}_i - \mathbf{c}_i$ in $[-\frac{1}{2}, \frac{1}{2})^2$.	pixel
n_{sub}	Effective integer phase resolution represented by subpixel_n ; every $n_{\text{sub}} \geq 1$ is permitted under the owner-approved upper-bin phase-lattice disposition.	1
$\mathbf{q}_i$	Two-axis phase-bin index in $\{0, \dots, n_{\text{sub}} - 1\}^2$.	index
$\hat{\phi}_i$	Midpoint representative of the exact half-open phase bin selected by $\mathbf{q}_i$; for even n_{sub} , exact zero phase selects $+1/(2n_{\text{sub}})$ on each applicable WCS axis.	pixel
β_i	Exact axis-1-fast linear phase index $q_{i1} + n_{\text{sub}}q_{i2}$.	index
$\mathcal{K}_{id}$	Logical coefficient-cache key (β_i, d_1, d_2) , equivalent to $(q_{i1}, q_{i2}, d_1, d_2)$.	identity
$\mathbf{M}$	Exact two-by-two affine tangent-plane pixel matrix of the target WCS, including axis order, signs, rotation, and handedness.	angle per pixel
Δ	Positive target-pixel angular scale satisfying $\mathbf{M}^T \mathbf{M} = \Delta^2 \mathbf{I}$ for the base-v0.1 square-pixel WCS class.	angle per pixel
$\mathcal{D}$	Finite destination pixel-center domain $\{1, \dots, N_x\} \times \{1, \dots, N_y\}$.	identity
h_a	Integer square-cache half-width $\lceil s_a(r_{\text{max}})_a / \Delta \rceil$.	pixel
$\mathbf{d}$	Integer cache offset in $\{-h_a, \dots, h_a\}^2$.	pixel
z_i	Exact finite transformed signal Z_i^{PTC} .	U
ω_i	Positive, finite, PTC-produced analysis/gridding coefficient from one exact registered family/version that permits SCI-JINC.	1 or U^{-2} , as declared by its family
κ_{ip}	Signed, dimensionless analytic JINC coefficient at the selected point phase. Finite positive, exact-zero, and finite negative values are distinct legitimate analytic results.	1

Symbol	Meaning and domain	Unit
w_{ip}	Complete signed contribution coefficient, $w_{ip} = \omega_i \kappa_{ip}$.	unit of ω_i
I_{ip}	Boolean membership after sample admission, exact AST association, rounded-center domain admission, local square geometry, and all pixel-local gates.	1
s_a	Explicit angular radial scale associated with stable array a ; finite and strictly positive in any numerical realization.	angle
r	Generic angular separation used by the analytic family.	angle
$\widehat{r}_{id}$	Exact discrete angular separation between the representative sample phase and destination pixel center for cache offset $\mathbf{d}$, from Equation (13).	angle
r'_a	Dimensionless radius r/s_a , finite and nonnegative.	1
$j_{1,1}$	Exact first positive zero $\min\{x > 0 : J_1(x) = 0\}$ of the first-order Bessel function.	1
$a_a, b_a, c_a, (r_{\max})_a$	Ordered, dimensionless analytic shape tuple for array a ; every member is finite and strictly positive in an authorized numerical realization.	1
θ_a	Complete kernel parameter identity $(s_a, a_a, b_a, c_a, (r_{\max})_a)$, including owner, source, version, units, array association, and lifecycle state.	mixed, as above
N_p	Signed signal numerator.	$U[\omega]$
C_p	Signed normalization; it may be finite and negative, but exact zero is never accepted.	$[\omega]$
Q_p	Quadratic accumulator; finite and strictly positive on accepted local support.	$[\omega]$
m_p	Normalized estimate N_p/C_p on accepted local support.	U
D_p	Absolute-term sum $\sum_i I_{ip} \omega_i \kappa_{ip} $, used only as construction state for conditioning.	$[\omega]$
ρ_p	Dimensionless cancellation statistic $ C_p /D_p$.	1
$\mathcal{P}_{\text{NA}}$	Exact owner-approved numerical-adequacy profile, including its observable, norm, domain, error treatment, exact inequality, uncertainty rule, lifecycle, provenance, and explicit exact-discrete-oracle-only claim boundary.	identity
$\mathcal{E}_p$	Exact numerical-adequacy certificate for pixel p under $\mathcal{P}_{\text{NA}}$; a passing certificate establishes only the declared finite-precision fidelity to the exact selected discrete oracle.	identity

Symbol	Meaning and domain	Unit
G_a	Compact bundle-level generative record for one observation and array; it identifies immutable parents, selections, joins, operator state, lifecycle, and completion without storing dense per-contribution provenance.	identity
A_{pi}	Fixed-state normalized linear operator row $I_{ip}\omega_i\kappa_{ip}/C_p$.	1
$f_{s,i}$	Effective processed-sample frequency, finite and strictly positive.	s^{-1}
$T_p^{(\kappa^2)}$	Required <code>jinc_coefficient_squared_time</code> coefficient-squared temporal-accounting accumulator.	s

In-memory indices are zero-based. Persisted FITS/WCS pixel coordinates and the mathematical coordinates in Equations (4)–(13) are one-based at integer pixel centers. Memory order never supplies axis sign, handedness, orientation, or wrapping.

D Normative definitions and boundary

SCI-JINC observation estimator.

A signed spatial-kernel transformer that consumes exact PTC signal occurrences and their exact AST coordinates and forms at most one complete JINC bundle for each admitted and requested stable array in one observation. It is a sibling alternative to ordinary SCI-MAP, not a downstream stage of SCI-MAP.

Exact occurrence.

The scientific identity that jointly fixes observation, detector occurrence and UID, stable RTC output sample, PTC product and application generation, PTC segment, stable array/group, time attribute, and immutable parent chain. Row position, nearest time, tolerance, shape, detector order, or numerical equality is not a substitute.

Exact AST association.

The relation binding z_i to the AST-owned `SCI-AST:rtc_output_grid_coordinates@1` continuous coordinate for the same processed sample realization in the exact target JINC WCS. Its data-model join is an engineering choice; its scientific identity is not.

JINC upstream-occurrence admission.

The canonical human-readable term for the decision that one exact upstream occurrence may be considered by JINC. Its immutable Registry identity is `SCI-JINC:jinc_map_contribution@1`; the identity remains fixed, but its name shall not be used as shorthand for arithmetic contribution. This decision neither authorizes rounded-center admission, sample-pixel membership, cancellation support, numerical contribution, nor bundle publication.

Rounded-center domain admission.

The JINC-owned occurrence-level gate applied after center rounding and before any footprint evaluation. The resolved rounded center must lie within the finite destination pixel domain. An outside center makes $I_{ip} = 0$ for every p , even when the square footprint would overlap the map.

Point phase.

The residual between the exact continuous sample coordinate and the exact rounded center of Equation (4). Equations (6)–(8) select a half-open bin and its midpoint representative independently on

each axis. The resulting kernel matrix is point evaluated and is not a pixel-area average. Under the owner-approved upper-bin disposition, every positive n_{sub} remains permitted. For even n_{sub} , exact zero phase selects the positive midpoint on each applicable WCS axis; no zero-phase special case replaces that representative. The positive-axis half-pixel center tie is likewise intentional.

Square support.

The fully populated integer-offset cache of Equation (12), with exact half-width $h_a = \lceil s_a(r_{\text{max}})_a / \Delta \rceil$. Every resolved square pixel is evaluated, including corners with radius larger than $(r_{\text{max}})_a$; there is no circular membership predicate.

Sample-pixel membership.

The Boolean proposition $I_{ip} = 1$ after sample admission, exact coordinate association, rounded-center domain admission, square placement, finite local geometry, and every pixel-local precondition. A contract-defined analytic zero is a member with zero contribution; outside support is nonmembership; a finite negative coefficient is an ordinary signed member.

Algebraic local support.

The exact-arithmetic state in which the required inputs and accumulators are finite, $Q_p > 0$, and $C_p \neq 0$. It establishes that N_p/C_p is algebraically defined; it does not certify a finite-precision realization.

Certified numerical-realization adequacy.

The state in which an exact owner-approved profile $\mathcal{P}_{\text{NA}}$ and a matching certificate $\mathcal{E}_p$ bind the scientific observable, absolute or relative form, reference norm, parameter/source/pixel domain, numerator and denominator error treatment, near-zero behavior, simultaneous or worst-case interpretation, exact pass/fail inequality, uncertainty, lifecycle, and provenance, and the certificate passes that inequality. No such profile or certificate is admitted or supplied by this source packet, so numerical-realization adequacy is presently typed unavailable rather than approximated by prose. A passing pair establishes only finite-precision fidelity to the exact selected discrete SCI-JINC oracle over its declared domain. It does not qualify the selected parameters, phase resolution, square truncation, scientific response, performance, validation, readiness, or production use.

Exact-source lock.

The base-v0.1 compatibility rule that binds the complete exact PTC boundary, AST boundary, admission-profile object, SCI-VAL binding, and JINC-specific source/profile successor snapshot bytes named by SCI-JINC-REQ-043. Ambient or merely current Registry state is not a substitute. Any changed bound object requires a versioned SCI-JINC successor; numerical or textual similarity cannot establish compatibility.

Complete observation bundle.

The atomic fixed-role product for one observation, stable array, exact JINC plan and realization, target WCS, and lifecycle generation. It contains exactly five numerical map-plane roles: N_p , C_p , Q_p , derived m_p with local support/validity, and $T_p^{(\kappa^2)}$. Its compact bundle-level generative record G_a is required information state at the existing plan/bundle provenance granularity; it is not a sixth product role, a dense per-contribution archive, or generalized provenance machinery.

Whole-bundle failure.

Failure to form or publish any required whole-product role. It suppresses the affected bundle without a partial role, placeholder, synthetic empty-array product, or completion override. Pixel-local invalid support is ordinary bundle content and is not whole-bundle failure.

Typed numerical unavailability.

The fail-closed state in which a numerical JINC route cannot be realized because an exact authorized TolTEC kernel parameter set or an exact registered, selected, realized, JINC-permitted PTC coefficient family/payload is absent. The same state applies when an exact owner-approved numerical-adequacy profile and matching certificate are required but absent. It is not repaired by a hidden default, unity coefficient, alternate family, ordinary MAP permission, inherited scale, implementation value, or approximate fidelity phrase.

Scientific destination identity.

The product identity formed from observation, stable array, exact JINC plan and realization, target WCS, product role, and lifecycle generation. Worker, thread, process, container, filename, vector position, and network order are not scientific identity.

E Canonical equations

E.1 Peak-normalized analytic family

Let J_1 be the first-order Bessel function of the first kind, and define its first positive zero exactly by

$$j_{1,1} = \min\{x > 0 : J_1(x) = 0\}. \quad (1)$$

The rounded decimal 3.831706 is only a nonnormative approximation to $j_{1,1}$. Define the collision-free peak-normalized function

$$\mathcal{J}(x) = \begin{cases} 2J_1(x)/x, & x \neq 0, \\ 1, & x = 0. \end{cases} \quad (2)$$

For $r'_a = r/s_a$, the generic SCI-JINC coefficient is

$$\kappa_a(r'_a; a_a, b_a, c_a, (r_{\max})_a) = \mathcal{J}\left(\frac{2\pi r'_a}{a_a}\right) \exp\left[-\left(\frac{2r'_a}{b_a}\right)^{c_a}\right] \mathcal{J}\left(\frac{j_{1,1}r'_a}{(r_{\max})_a}\right). \quad (3)$$

Equation (3) has $\kappa_a(0) = 1$. a_a scales the first JINC argument, b_a scales the generalized-exponential envelope, c_a is its exponent, and $(r_{\max})_a$ places the first zero of the second JINC factor. Analytic zeros and finite signed lobes are retained. The square-cache rule, rather than a radial cutoff, determines membership.

E.2 Exact discrete point-phase and WCS operator

Let the exact AST continuous FITS pixel coordinate in the target JINC WCS be $\mathbf{u}_i = (x_i, y_i)$. Integer FITS coordinates denote pixel centers. The exact rounded cache center is

$$\mathbf{c}_i = (\lfloor x_i + \frac{1}{2} \rfloor, \lfloor y_i + \frac{1}{2} \rfloor). \quad (4)$$

Thus an exact half-pixel tie rounds toward the positive coordinate axis. The scientific owner retains this directional convention as part of the selected discrete operator. The residual phase is

$$\phi_i = \mathbf{u}_i - \mathbf{c}_i \in [-\frac{1}{2}, \frac{1}{2})^2. \quad (5)$$

For $n_{\text{sub}} \geq 1$, each axis uses the exact half-open bins

$$B_q = \left[-\frac{1}{2} + \frac{q}{n_{\text{sub}}}, -\frac{1}{2} + \frac{q+1}{n_{\text{sub}}}\right), \quad q \in \{0, \dots, n_{\text{sub}} - 1\}, \quad (6)$$

$$q(\phi) = \lfloor n_{\text{sub}}(\phi + \frac{1}{2}) \rfloor, \quad (7)$$

$$\hat{\phi}_q = -\frac{1}{2} + \frac{q + \frac{1}{2}}{n_{\text{sub}}}. \quad (8)$$

The two axis indices form $\mathbf{q}_i$, and their midpoint representatives form $\hat{\phi}_i$. Every positive integer n_{sub} is permitted under the owner-approved upper-bin phase-lattice disposition. The half-open upper-bin

convention is observable: when n_{sub} is even, an exact $\phi = 0$ selects $q = n_{\text{sub}}/2$ and $\hat{\phi} = +1/(2n_{\text{sub}})$, with the sign interpreted in the exact target-WCS axis orientation. No special case substitutes a zero representative. The exact logical cache mapping is

$$\beta_i = q_{i1} + n_{\text{sub}}q_{i2}, \quad \mathcal{K}_{i\mathbf{d}} = (\beta_i, d_1, d_2), \quad (9)$$

equivalently $(q_{i1}, q_{i2}, d_1, d_2)$ with axis 1 varying fastest. Physical storage may use any proven value-preserving bijection of this logical key.

Base v0.1 requires an affine tangent-plane target WCS with exact pixel matrix $\mathbf{M}$ satisfying

$$\mathbf{M}^\top \mathbf{M} = \Delta^2 \mathbf{I}, \quad \Delta > 0. \quad (10)$$

The exact $\mathbf{M}$ retains axis order, signs, rotation, and handedness; Equation (10) prohibits representing an anisotropic or skewed metric by a scalar pixel size. With finite destination domain $\mathcal{D} = \{1, \dots, N_x\} \times \{1, \dots, N_y\}$, define

$$h_a = \left\lceil \frac{s_a(r_{\text{max}})_a}{\Delta} \right\rceil, \quad (11)$$

$$\mathbf{d} \in \{-h_a, \dots, h_a\}^2, \quad \mathbf{p} = \mathbf{c}_i + \mathbf{d}. \quad (12)$$

For every retained square offset, the exact discrete point-phase radius is

$$\hat{r}_{i\mathbf{d}} = \left\| \mathbf{M}(\mathbf{d} - \hat{\phi}_i) \right\|_2, \quad \hat{r}'_{i\mathbf{d}} = \hat{r}_{i\mathbf{d}}/s_a, \quad (13)$$

and κ_{ip} is Equation (3) evaluated at $\hat{r}'_{i\mathbf{d}}$. The occurrence passes the center-domain gate only when $\mathbf{c}_i \in \mathcal{D}$; a square pixel is retained only when $\mathbf{p} \in \mathcal{D}$. These integer-domain predicates, rather than footprint overlap or a radial cutoff, determine geometry membership.

For an occurrence at an exact integer pixel center, both residual-phase components are zero. When n_{sub} is even, the owner-approved upper-bin rule and Equation (10) therefore give, at the nominal central offset $\mathbf{d} = \mathbf{0}$,

$$\begin{aligned} \hat{r}_{i\mathbf{0}} &= \frac{\Delta}{\sqrt{2} n_{\text{sub}}}, \\ \kappa_{i,\mathbf{c}_i} &= \kappa_a \left(\frac{\Delta}{\sqrt{2} n_{\text{sub}} s_a}; a_a, b_a, c_a, (r_{\text{max}})_a \right). \end{aligned}$$

The nominal central cache element is thus evaluated at this exact nonzero radius and shall not be replaced by the zero-radius limit $\kappa_a(0) = 1$. For odd n_{sub} , exact zero phase has midpoint representative zero and the nominal central offset evaluates the analytic limit at zero.

E.3 Signed estimator

For the exact occurrence population of one bundle,

$$w_{ip} = \omega_i \kappa_{ip}, \quad (14)$$

$$N_p = \sum_i I_{ip} \omega_i \kappa_{ip} z_i, \quad (15)$$

$$C_p = \sum_i I_{ip} \omega_i \kappa_{ip}, \quad (16)$$

$$Q_p = \sum_i I_{ip} \omega_i \kappa_{ip}^2, \quad (17)$$

$$m_p = \frac{N_p}{C_p}, \quad (18)$$

$$A_{pi} = \frac{I_{ip} \omega_i \kappa_{ip}}{C_p}. \quad (19)$$

All coupled accumulator terms for one contribution use the same I_{ip} and the same realized κ_{ip} identity. The normalization is signed. A finite negative C_p is admissible; exact $C_p = 0$ is not a zero-sky result.

E.4 Conditioning and local support

Define construction-state quantities

$$D_p = \sum_i I_{ip} |\omega_i \kappa_{ip}|, \quad (20)$$

$$\rho_p = \frac{|C_p|}{D_p}, \quad D_p > 0. \quad (21)$$

For exact arithmetic, $0 \leq \rho_p \leq 1$. Accepted local support requires the algebraic predicate

$$\text{AlgSupport}_p = [\text{required inputs and } N_p, C_p, Q_p \text{ are finite}] \wedge [Q_p > 0] \wedge [C_p \neq 0]. \quad (22)$$

A finite-precision numerical realization additionally requires

$$\text{NumAdeq}_p = [\mathcal{E}_p \text{ is a valid certificate under } \mathcal{P}_{\text{NA}}] \wedge [\mathcal{E}_p \models \mathcal{P}_{\text{NA}}]. \quad (23)$$

The exact profile supplies the observable, norm, domain, error propagation, near-zero rule, simultaneous or worst-case interpretation, and pass/fail inequality. No such owner-approved profile or matching certificate is admitted or supplied by this source packet, so Equation (23) is presently typed unavailable. In particular, no approximate fidelity scale promotes a near-cancellation pixel. D_p remains construction state rather than a numerical map-plane role; any future certificate is referenced by the compact generative record G_a . A passing $\mathcal{P}_{\text{NA}}/\mathcal{E}_p$ pair establishes only finite-precision fidelity to the exact selected discrete SCI-JINC oracle over the profile's declared domain. It does not establish scientific adequacy of s_a , a_a , b_a , c_a , $(r_{\text{max}})_a$, n_{sub} , or square truncation; source-response or beam fidelity; photometric or astrometric performance; optimality; observational validation; response/covariance fidelity; readiness; or production authority. Qualification of a TolTEC parameter set is a separate scientific and observational program.

E.5 Coefficient-squared temporal accounting

The sole authorized base-v0.1 coefficient-squared temporal-accounting product is `jinc_coefficient_squared_time`, defined by

$$T_p^{(\kappa^2)} = \sum_i I_{ip} \frac{\kappa_{ip}^2}{f_{s,i}}. \quad (24)$$

It is nonnegative and has unit seconds. The squared object is κ_{ip} , not ω_i or w_{ip} .

E.6 Conditional mathematics retained for future science

The following identities do not create base-v0.1 products. For an exact available realized PTC-grid source-template response, the fixed-state response is

$$\mathbf{R}_{\text{JINC, fixed}} = \mathbf{A}_{\text{JINC}} \mathbf{H}_{\text{PTC, fixed}}. \quad (25)$$

The exact signal membership is used and the already realized PTC companion receives the JINC operator exactly once. A PTC full-procedure finite difference with JINC held fixed, a JINC re-resolved procedure response, and a whole-chain response are distinct families and must not be substituted for Equation (25) or for one another.

If and only if an exact PTC family establishes $\omega_i = \text{Var}(z_i)^{-1}$ for mutually independent admitted occurrences,

$$\text{Var}_{\text{formal}}(m_p) = \frac{Q_p}{C_p^2}, \quad (26)$$

$$W_{\text{formal}, p} = \frac{C_p^2}{Q_p}, \quad (27)$$

$$\text{Cov}(m_p, m_{p'}) = \frac{\sum_i I_{ip} I_{ip'} \omega_i \kappa_{ip} \kappa_{ip'}}{C_p C_{p'}}. \quad (28)$$

For a general available upstream covariance,

$$\mathbf{C}_{\text{JINC}} = \mathbf{A}_{\text{JINC}} \mathbf{C}_{\text{PTC}} \mathbf{A}_{\text{JINC}}^T. \quad (29)$$

Inverse-square units alone do not establish inverse variance, independence, precision, covariance authority, or significance. A covariance state, if a future use authorizes one, is exactly one of complete numerical, partial, symbolic, summarized, lineage-resolvable, or unavailable and declares every omitted term. Missing covariance is never zero or independence. Response, variance, formal-weight, and covariance companions remain deferred until a concrete scientific use separately authorizes them.

F Assumptions, validity classes, and explicit unavailability

Table 9: Normative assumptions and boundary states.

ID	Statement
SCI-JINC-ASM-001	The contract concerns one observation at a time and a nonpolarimetric total-intensity-equivalent transformed signal. It does not authorize a physical Stokes interpretation from a component token.
SCI-JINC-ASM-002	PTC and AST meanings, identities, validity facts, causes, uncertainty, and immutable ancestry are producer-owned facts. JINC uses them without reconstructing or redefining them.
SCI-JINC-ASM-003	An admitted ω_i is finite and strictly positive and comes from one exact PTC family/version that explicitly permits SCI-JINC. No registered family or realized payload is admitted or supplied by this source packet.
SCI-JINC-ASM-004	Equations (25)–(29) are conditional identities only. Their assumptions are not inferred from units, names, or another consumer’s permission.
SCI-JINC-ASM-005	The analytic family and parameter meanings are approved, but no TolTEC numerical realization of s_a , a_a , b_a , c_a , $(r_{\max})_a$, target-WCS metric, pixel scale, or <code>subpixel_n</code> is scientifically admitted, supplied, or recommended by this source packet.
SCI-JINC-ASM-006	Point-phase evaluation and fully populated square support are the scientific operator. The owner-approved upper-bin phase lattice admits every positive n_{sub} , retains the positive midpoint for exact zero phase when n_{sub} is even, and retains positive-axis half-pixel center ties. Pixel-area integration, a zero-phase special case, a circular cutoff, footprint-overlap admission, and JINC-then-crop equivalence are not alternative branches.
SCI-JINC-ASM-007	Numerical-realization adequacy requires one exact owner-approved $\mathcal{P}_{\text{NA}}$ and a matching certificate $\mathcal{E}_p$. Neither is admitted or supplied by this source packet. Finite arithmetic, analytic-function, phase/cache, and accumulation errors therefore have no authorized approximate substitute, and numerical support is typed unavailable even though exact-arithmetic algebraic support is defined. A future passing profile/certificate pair establishes only finite-precision fidelity to its exact selected discrete oracle; it does not qualify TolTEC parameters, phase resolution, square truncation, response, performance, validation, readiness, or production use.
SCI-JINC-ASM-008	Same-observation chunking may be an engineering realization of one bundle, but a chunk is not a scientific product identity. Cross-observation combination is outside base v0.1.
SCI-JINC-ASM-009	This Stage B draft has inspected no implementation candidate, configuration, product, test, audit, validation, reduction, or production evidence.

Every missing fact material to this draft is classified in Table 10.

Table 10: Missing-fact classification.

Class	Fact	Consequence
Normatively specified in this draft	Generic analytic identity, signed estimator, exact point-phase square operator, center-domain gate, algebraic support, fixed five-plane bundle, grouping, and ownership boundary.	These are binding draft content in v0.1 but are not implementation or validation claims.
Not admitted or supplied by this source packet	A registered, selected, and realized JINC-permitted PTC coefficient family and payload.	No numerical bundle is formed without it; no unity or alternate-family fallback.
Not admitted or supplied by this source packet	A scientifically authorized TolTEC array-associated numerical parameter set.	No numerical bundle is formed; no inherited scale, shape, extent, or mode-dependent value is used.
Not admitted or supplied by this source packet	An exact owner-approved numerical-adequacy profile and a matching lifecycle-bound certificate.	Algebraic support remains defined, but numerical-realization adequacy, including near-cancellation support, is typed unavailable.
Implementation unassessed	Representation, interfaces, algorithms, storage, parallel realization, and implementation behavior.	No conformity or representation-fidelity claim.
Validation not performed	Algebraic execution, numerical adequacy, observational response, covariance, photometry, performance, readiness, and production operation.	No achieved result, readiness, or production claim.

Response, covariance, formal weight, physical exposure, cross-observation coaddition, standalone support or availability, diagnostics, dense per-contribution provenance, and generalized provenance products are deferred or outside base v0.1, rather than silently missing members of the fixed five-plane bundle. The compact replay record G_a is required bundle-level information state, not an additional product role or generalized provenance framework.

G Binding scientific requirements

The key words *shall*, *shall not*, *required*, and *unavailable* are normative. These requirements define scientific conformance; they do not assert that any implementation satisfies them.

SCI-JINC-REQ-001. SCI-JINC shall define a signed spatial-kernel estimator for one observation and stable array. It shall consume exact PTC and AST parents directly, consume no SCI-MAP product, and inherit no ordinary SCI-MAP normalization, support, exposure, validity, response, covariance, product, or coaddition rule by analogy.

SCI-JINC-REQ-002. Every candidate contribution shall bind one exact occurrence and the SCI-AST: `rtc_output_grid_coordinates@1` continuous coordinate under SCI-AST_TO_SCI-JINC v0.1/r0.2, associated with the same processed sample realization in the exact target JINC WCS. Row order, detector order, nearest time, tolerance, shape, cardinality, or numerical coordinate equality shall not establish that association.

SCI-JINC-REQ-003. The transformed signal z_i shall retain its exact PTC quantity, unit, availability, application generation, retention state, immutable parents, nonrestored additive reference, removed-component and null-space state, response/covariance producer facts when present, and producer causes. SCI-JINC shall spatially transform that quantity without creating calibration, conditioning, astrometry, empirical noise, or source interpretation.

SCI-JINC-REQ-004. Every ω_i shall come from one exact family/version in the single PTC-owned positive analysis/gridding coefficient registry. The family shall explicitly permit SCI-JINC, and its requested, effective, observation-resolved, and realized identities, payload, QC, normalization, support, provenance, covariance meaning, uncertainty, and prohibited meanings shall remain bound. The accepted request shall select an explicitly permitted family, or one exact owner-authorized versioned mode policy shall select it; no ambient or implementation default is authorized.

SCI-JINC-REQ-005. A numerical route shall fail closed when selection by the accepted request or an exact owner-authorized versioned mode policy is missing, the family is unregistered, JINC permission is absent, or the payload/QC/identity is absent, unavailable, incompatible, duplicate, or ambiguous. SCI-JINC shall not infer unity, another family, ordinary MAP permission, **sens**, loading, scatter, inverse variance, precision, or significance.

SCI-JINC-REQ-006. After the producer-owned coefficient decision, SCI-JINC shall admit only a finite strictly positive ω_i . Zero, negative, non-finite, or unrepresentable upstream coefficient values shall not be deposited and shall retain the exact profile-defined disposition and cause.

SCI-JINC-REQ-007. Every numerical kernel route shall bind one exact scientifically authorized array-associated parameter-set identity containing finite positive s_a , a_a , b_a , c_a , $(r_{\max})_a$, effective integer `subpixel_n` ≥ 1 with no parity restriction under the owner-approved upper-bin phase-lattice disposition, and compatibility with the exact target-WCS metric. Base v0.1 requires an affine tangent-plane WCS whose exact pixel matrix obeys Equation (10). A scalar pixel size shall not represent an anisotropic or skewed geometry. The exact frame, tangent projection, reference point, axis order and signs, rotation, handedness, angular unit, dimensions, and finite nonwrapping topology shall remain bound. Missing or incompatible authority shall make the affected route unavailable without fallback.

SCI-JINC-REQ-008. The peak-normalized analytic primitive shall be $\mathcal{J}(x) = 2J_1(x)/x$ for $x \neq 0$ with the removable limit $\mathcal{J}(0) = 1$, as in Equation (2).

SCI-JINC-REQ-009. The analytic coefficient shall follow Equation (3), including its fixed parameter ordering and $r'_a = r/s_a$. It shall have peak one, preserve analytic zeros, and preserve every finite positive and negative lobe of the complete product. The second-factor zero shall use the exact root $j_{1,1} = \min\{x > 0 : J_1(x) = 0\}$ of Equation (1); a rounded decimal shall not be its normative definition.

SCI-JINC-REQ-010. The sample center, half-pixel tie, residual-phase domain, half-open phase bins, phase-to-cache index, and midpoint representatives shall be exactly Equations (4)–(9), independently on the two exact WCS axes. Every positive integer n_{sub} is permitted. For even n_{sub} , exact zero phase shall select the positive representative $+1/(2n_{\text{sub}})$ under the bound WCS axis orientation. The selected nominal central offset shall therefore use the exact even-lattice center formulas following Equation (13),

not the zero-radius limit. The positive-axis half-pixel center tie and upper-bin phase convention are scientific-owner-intended parts of the operator. The selected kernel matrix shall be point evaluated and shall not be replaced by pixel-area integration.

SCI-JINC-REQ-011. Center rounding, tie handling, phase intervals, bin edges, representatives, phase-to-cache indexing, WCS metric, cache extent, destination evaluation point, and edge membership are fixed mathematical parts of the operator and shall not be implementation choices. Bessel evaluation, summation algorithm, cache storage, thread order, and parallel reduction may be engineering choices only when they realize the exact operator and carry the numerical-adequacy state required by Equations (22)–(23).

SCI-JINC-REQ-012. $(r_{\max})_a$ shall place the exact first zero of the second JINC factor and shall fix $h_a = \lceil s_a(r_{\max})_a / \Delta \rceil$. The offsets shall be exactly $\{-h_a, \dots, h_a\}^2$, and each coefficient shall use the destination pixel-center and representative-phase radius of Equation (13). Every resolved square pixel, including a corner with radius greater than $(r_{\max})_a$, shall be evaluated; no circular cutoff or radial membership predicate shall be applied.

SCI-JINC-REQ-013. Before any footprint evaluation, the exact rounded cache center of Equation (4) shall lie in $\mathcal{D} = \{1, \dots, N_x\} \times \{1, \dots, N_y\}$. An outside center shall set $I_{ip} = 0$ for every pixel, even when its square would overlap the map; footprint-overlap admission is prohibited.

SCI-JINC-REQ-014. For an admitted in-map center, the exact square pixel $\mathbf{p} = \mathbf{c}_i + \mathbf{d}$ shall be retained if and only if $\mathbf{p} \in \mathcal{D}$ and every other local gate passes; square pixels outside the finite map shall be absent. The route shall not wrap, reflect, complete the footprint, substitute an interior normalizer, renormalize the edge, or require JINC-then-crop equivalence.

SCI-JINC-REQ-015. Every evaluated κ_{ip} shall be finite. A finite negative coefficient is normal and shall retain its sign; an analytic zero is a contract-defined zero contribution distinct from outside support; a non-finite coefficient shall fail the affected required bundle before deposition rather than be clipped, skipped, or zero-filled.

SCI-JINC-REQ-016. All coupled accumulator terms for one contribution shall use the same exact occurrence, AST association, sample-admission decision, rounded-center decision, sample-pixel membership, phase, and κ_{ip} identity.

SCI-JINC-REQ-017. The required accumulators shall be distinct and shall obey Equations (15)–(17). Negative lobes shall keep their sign in N_p and C_p and enter Q_p quadratically.

SCI-JINC-REQ-018. The published signal estimate shall be $m_p = N_p/C_p$ and the fixed-state normalized operator shall be Equation (19). No Q_p denominator or other local measurement model shall be substituted.

SCI-JINC-REQ-019. Algebraic local support shall be exactly Equation (22): finite admitted inputs, finite N_p , C_p , and Q_p , $Q_p > 0$, and $C_p \neq 0$. This state shall remain distinct from certified finite-precision numerical adequacy. A rejected pixel shall carry local invalid support in the `jinc_map` role and shall not receive a finite substitute.

SCI-JINC-REQ-020. A finite negative C_p shall be admissible. Exact $C_p = 0$ shall be locally invalid and unavailable, never interpreted as zero sky, infinity, or a valid non-finite map value.

SCI-JINC-REQ-021. SCI-JINC shall retain the dimensionless conditioning statistic ρ_p of Equation (21). It is an indicator and shall not itself promote numerical support. A finite-precision result shall be

numerically supported only when an exact owner-approved $\mathcal{P}_{\text{NA}}$ and matching $\mathcal{E}_p$ satisfy Equation (23); otherwise numerical adequacy, including near-cancellation support, shall be typed unavailable.

SCI-JINC-REQ-022. No ambient universal ρ_p cutoff, unit-bearing C_p or Q_p floor, contributor-count/machine-epsilon formula, prescribed summation algorithm, fixed reduction order, bitwise reproducibility, approximate fidelity phrase, or implementation default shall be treated as scientific authority. A future exact numerical-adequacy profile may define only its explicitly bounded domain-specific predicate.

SCI-JINC-REQ-023. Every claimed numerical-support result shall bind one exact owner-approved numerical-adequacy profile and matching certificate. Together they shall identify the scientific observable; absolute or relative form; reference magnitude or norm; parameter, source, and pixel domain; numerator and denominator error treatment; near-zero behavior; simultaneous or worst-case interpretation; exact pass/fail inequality; uncertainty; lifecycle; and provenance. No such profile or certificate is admitted or supplied by this source packet, so no numerical-support result is presently authorized. A passing pair shall establish only finite-precision fidelity to the exact selected discrete SCI-JINC oracle over its declared domain. It shall not establish scientific adequacy of the kernel parameters, n_{sub} , or square truncation; source-response or beam fidelity; photometric or astrometric performance; optimality; observational validation; response/covariance fidelity; readiness; or production authority.

SCI-JINC-REQ-024. The unit relations in Table 8 and the two allowed ω_i unit classes shall be preserved. Common positive scaling of all ω_i shall leave m_p , ρ_p , and A_{pi} invariant; signal-unit rescaling shall transform the estimator consistently.

SCI-JINC-REQ-025. Equations (26)–(29) may receive a formal statistical interpretation only when the exact PTC family and covariance assumptions establish it. Inverse-square units or historical labels alone shall not establish variance, precision, covariance, or significance. Any future covariance state shall be exactly typed as complete numerical, partial, symbolic, summarized, lineage-resolvable, or unavailable, shall declare omitted terms, and shall never interpret missing covariance as zero or independence.

SCI-JINC-REQ-026. The required coefficient-squared temporal-accounting product `jinc_coefficient_squared_time` shall be exactly Equation (24). It shall square κ_{ip} , not ω_i or w_{ip} , and shall have unit seconds.

SCI-JINC-REQ-027. `jinc_coefficient_squared_time` shall be described only as method-specific coefficient-squared temporal accounting. It shall not be interpreted as physical acquired exposure, valid-original exposure, complete temporal support, hits, normalized influence, white-noise-equivalent time, precision, validity, confidence, or significance.

SCI-JINC-REQ-028. Every produced base-v0.1 bundle shall contain exactly five required numerical map-plane roles: `jinc_signal_numerator`, `jinc_signed_normalization`, `jinc_quadratic_accumulator`, derived `jinc_map` with local support/validity, and `jinc_coefficient_squared_time`. The compact bundle-level generative record required by SCI-JINC-REQ-044 shall be information state at the existing plan/bundle provenance granularity and shall not be a sixth product role.

SCI-JINC-REQ-029. Failure to form or publish any required whole-product role shall suppress the affected bundle atomically. Pixel-local zero, insufficient, cancelled, or otherwise invalid support shall remain ordinary bundle content and shall not suppress an otherwise complete role or create a role-availability object.

SCI-JINC-REQ-030. One observation may produce zero through three independent bundles, with at

most one for each stable array admitted and requested under the exact JINC realization. A missing, unavailable, or unrequested array shall create no placeholder or empty-array product and shall not invalidate another array's bundle.

SCI-JINC-REQ-031. Each bundle shall be independently identified by observation, stable array, exact JINC plan and realization, target WCS, product role, and lifecycle generation. Its compact generative record G_a shall bind those identities without making a worker, filename, or dense contribution archive part of scientific identity. Contributions with different array or destination identities shall never merge.

SCI-JINC-REQ-032. The complete destination identity shall be resolved before allocation, numerical mutation, output side effects, or parallel execution. Ambiguous, duplicate, or aliased destination ownership shall fail the affected bundle without selecting a winner or publishing a partial result.

SCI-JINC-REQ-033. Same-observation samples or chunks may accumulate incrementally only when observation, stable array, JINC plan and realization, target WCS, admission/parameter/coefficient state, destination, and lifecycle generation are identical and are replayable from the same G_a . Chunk, worker, process, container, filename, and vector position shall not create product identity.

SCI-JINC-REQ-034. Base v0.1 shall authorize no cross-observation combination. Any future JINC coadd shall require a separately authorized boundary over complete observation bundles and shall not infer accumulator addition, normalized-map combination, or ordinary SCI-MAP coaddition.

SCI-JINC-REQ-035. JINC upstream-occurrence admission, whose immutable Registry identity is `SCI-JINC:jinc_map_contribution@1`, shall remain distinct from PTC signal availability, PTC retention, coefficient availability/QC, AST coordinate validity and association, finite signal, finite positive ω_i , parameter availability, rounded-center admission, pixel support, finite κ_{ip} , cancellation, numerical adequacy, arithmetic contribution, and final bundle validity. The immutable Registry identity shall not routinely be described as a “map contribution” decision; its canonical human-readable meaning is only that one exact upstream occurrence may be considered by JINC.

SCI-JINC-REQ-036. Only a fact named by an exact restriction shall control JINC upstream-occurrence admission. Producer causes and transitive influence shall be preserved, but a generic flag or cause shall have no universal veto. Outside support, an analytic zero, or an outside rounded center shall be ordinary no-contribution; a finite negative lobe shall not create a cause. Profile pass shall not imply rounded-center admission, pixel membership, cancellation support, numerical contribution, or bundle publication.

SCI-JINC-REQ-037. ALIGN/AST, RTC, CAL, PTC, and VAL shall retain the meanings they own. SCI-JINC shall own only its admission use, local geometry, point phase, square support, signed coefficient, normalization, conditioning, fixed accumulators, local support, grouping, destination, and atomic bundle publication. The exact PTC r0.3 and AST r0.2 boundaries, immutable admission profile, and JINC-specific SCI-VAL successor snapshots shall control compatibility under the exact-source lock of SCI-JINC-REQ-043; SCI-JINC shall not reconstruct their producer facts.

SCI-JINC-REQ-038. Base v0.1 shall publish no response, variance, formal-weight, covariance, empirical-noise, or significance product. The conditional mathematics in Section E.6 shall remain preserved for a future concrete scientific use without being promoted to a present role. Fixed-state JINC response, PTC full-procedure response with JINC fixed, JINC re-resolved procedure response, and whole-chain response shall remain distinct; an already realized PTC-grid response shall receive the JINC operator exactly once and shall use exact signal membership.

SCI-JINC-REQ-039. Base v0.1 shall publish no physical-exposure product. Any future exposure role shall require separate authority for exact original- occurrence lineage and semantics and shall not reinterpret $T_p^{(\kappa^2)}$.

SCI-JINC-REQ-040. Base v0.1 shall create no standalone support or availability role, generic optional or conditionally required role mechanism, detailed operational-cause archive, diagnostic role, generalized provenance product, or per-pixel or per-contribution provenance system. This prohibition shall not remove the compact bundle-level generative record required by SCI-JINC-REQ-044.

SCI-JINC-REQ-041. No TolTEC numerical parameter realization, recommendation, optimum, hidden default, or fallback is authorized by v0.1. In particular, an inherited 45 m denominator, inherited shape values, or mode-dependent $r_{\max}$ values shall not be treated as scientific authority.

SCI-JINC-REQ-042. A future conformance assessment shall exercise every prediction in Section H, bind any exact owner-approved numerical- adequacy profile and certificate rather than an approximate acceptance phrase, and separate algebraic correctness from implementation conformity, representation/response fidelity, numerical and observational validation, achieved performance, readiness, and production authorization. A passing profile/certificate pair shall be reported only as exact-discrete-oracle finite-precision fidelity under SCI-JINC-REQ-023. This Stage B draft establishes none of those later claims.

SCI-JINC-REQ-043. Every claimed SCI-JINC contract realization shall bind byte identically to SCI-PTC_TO_SCI-JINC v0.1/r0.3 (SHA-256 5769d413460e931745e0d401ea432b12d1077c15466247c49caa71b997d4ab1e), SCI-AST_TO_SCI-JINC v0.1/r0.2 (SHA-256 efffa7059b59c89793fa1d523fb3bb48235f1ab55f7d55060af1600cbfd470a5), the SCI-JINC:jinc_map_contribution@1 profile (SHA-256 2db95da7e5d1b980df79993907d45ac0ababc3aa05c189bfb62dcf04ff2c2e8a), and the SCI-VAL binding record (SHA-256 ee8f20db5febdb51e39f7157449d6c2d03a0d17058605dd5531e9ab5ca439e30). The bound SCI-VAL source-register successor shall be SCI-VAL_SOURCE_BINDING_REGISTER version v0.1/r0.3-map-r0.7.1-jinc-stage-a-q002-2026-08-28 with SHA-256 0e7ca29ee2e9cd02fb1b76cf87cc64fce6164407a7801f9b9a105ca646317e88, and the profile-registry successor shall be SCI-VAL_PROFILE_REGISTRY version v0.1/r0.3-map-r0.7.1-jinc-stage-a-q002-2026-08-28 with SHA-256 4b9a1ebecfc847c83b59da772afd9b031ab1830e8feb12d1a47f70ce5a1110. The SCI-VAL successor identities and hashes shall be process bindings of the exact source/profile bytes only; they shall create no SCI-JINC scientific content, PTC coefficient family, TolTEC parameter set, or numerical-adequacy profile. Base v0.1 uses an exact-source-lock lifecycle: the complete exact files named above, including the JINC-stage-A-Q002 source/profile successor snapshots, are the bound objects. No ambient “current Registry” or nearby row/profile is consulted, and an unrelated Registry state does not replace these snapshot bytes. Any changed bound byte, identity, version, compatibility, or supersession fact shall make the affected route unavailable and requires a versioned SCI-JINC successor before use.

SCI-JINC-REQ-044. Every complete bundle shall contain one compact generative record G_a sufficient to replay its exact input population and immutable PTC/AST joins; admission evaluation; coefficient family, selection, generation, and payload identity; parameter-set identity; target WCS and metric; center, phase, bin, representative, cache, square, and edge rules; cancellation and numerical- adequacy state; destination identity; requested, effective, resolved, and realized lifecycle; the owner-approved upper-bin phase-lattice and positive-axis half-tie dispositions; the exact-source-lock identities; every required-role state; failure; and completion. The record shall use immutable references and bounded plan/bundle information state, not dense per-contribution provenance or a new product role.

H Falsifiable predictions and edge cases

These predictions are normative tests for a future authorized conformance program. Listing them is not evidence that they have been executed or passed.

Table 11: Stable SCI-JINC predictions.

Prediction	Falsifiable expected result
SCI-JINC-PRED-001	A constant finite input $z_i = z_0$ returns $m_p = z_0$ on every accepted local support, including a cropped edge pixel.
SCI-JINC-PRED-002	One admitted contributor with nonzero finite κ_{ip} returns $m_p = z_i$, whether the coefficient is positive or negative.
SCI-JINC-PRED-003	A hand-computed two-contributor case with unequal z_i , ω_i , and signed κ_{ip} matches the separate N_p , C_p , Q_p , m_p , and $T_p^{(\kappa^2)}$ equations.
SCI-JINC-PRED-004	Multiplying every ω_i by the same positive factor scales N_p , C_p , and Q_p together and leaves m_p , ρ_p , and A_{pi} unchanged.
SCI-JINC-PRED-005	Under a true inverse-variance family, rescaling $z_i \mapsto \lambda z_i$ and $\omega_i \mapsto \omega_i/\lambda^2$ produces $m_p \mapsto \lambda m_p$ and conditional formal weight $W_{\text{formal},p} \mapsto W_{\text{formal},p}/\lambda^2$.
SCI-JINC-PRED-006	The analytic evaluator returns $\kappa_a(0) = 1$ through the removable limit, not a non-finite value or an implementation-dependent substitute.
SCI-JINC-PRED-007	At $r'_a = (r_{\max})_a$, the exact identity $J_1(j_{1,1}) = 0$ makes the second JINC factor zero. The resulting analytic $\kappa_{ip} = 0$ changes none of N_p , C_p , Q_p , or $T_p^{(\kappa^2)}$ while remaining distinguishable from outside square support; a rounded decimal is not the root oracle.
SCI-JINC-PRED-008	A finite negative lobe retains its sign in N_p and C_p , enters Q_p and $T_p^{(\kappa^2)}$ quadratically, and is not clipped, absolutized, omitted, or flagged solely for being negative.
SCI-JINC-PRED-009	Exact cancellation $C_p = 0$ yields local invalid support and no finite substitute, even when N_p and Q_p are finite.
SCI-JINC-PRED-010	A finite negative C_p with otherwise adequate support yields the signed normalized estimate and is not rejected merely for the denominator sign.
SCI-JINC-PRED-011	A near-cancellation construction is numerically eligible only when a certificate $\mathcal{E}_p$ passes the exact inequality and complete domain rules of the bound owner-approved $\mathcal{P}_{\text{NA}}$; ρ_p alone neither promotes nor rejects it. A pass establishes only finite-precision fidelity to the exact selected discrete oracle on that domain and establishes none of the parameter, truncation, response, beam, photometric, astrometric, optimality, validation, fidelity, readiness, or production claims excluded by SCI-JINC-REQ-023.
SCI-JINC-PRED-012	With no exact owner-approved $\mathcal{P}_{\text{NA}}$ and matching $\mathcal{E}_p$, numerical adequacy is typed unavailable, even when algebraic support holds and finite arithmetic happens to return a nonzero C_p .

Prediction	Falsifiable expected result
SCI-JINC-PRED-013	A non-finite required signal, geometry, coefficient evaluation, accumulator, or sample frequency cannot be masked by multiplication by zero or silently skipped; the exact local or whole-bundle failure rule applies.
SCI-JINC-PRED-014	Square-cache corners whose radius exceeds $(r_{\max})_a$ are evaluated by Equation (3) and may contribute.
SCI-JINC-PRED-015	Two points on opposite sides of a hypothetical circular $r_{\max}$ boundary but inside the same resolved square are both governed by square membership; no radial cutoff appears.
SCI-JINC-PRED-016	Effective <code>subpixel_n</code> =1 selects bin $q = 0$ and midpoint representative $\hat{\phi} = 0$ on each axis for every residual phase; it selects one point-phase matrix and does not become a pixel-area average.
SCI-JINC-PRED-017	Coordinates immediately around every half-pixel tie and phase-bin edge select the exact centers, half-open bins, indices, and midpoint representatives of Equations (4)–(8). For even n_{sub} , exact $\phi = 0$ selects the positive representative $+1/(2n_{\text{sub}})$ in the bound WCS axis orientation. With zero phase on both axes and $\mathbf{d} = \mathbf{0}$, the exact radius is $\Delta/(\sqrt{2}n_{\text{sub}})$ and the coefficient is the exact even-lattice center formula following Equation (13), not an imposed $\kappa_a(0) = 1$. An exact positive half-pixel coordinate tie selects the larger center and residual phase $-1/2$. No parity restriction, zero-phase special case, or alternate center-tie rule is permitted.
SCI-JINC-PRED-018	The exact rounded center immediately outside each map edge or corner changes none of the four accumulators for any pixel, even when its square footprint overlaps the map.
SCI-JINC-PRED-019	A rounded center immediately inside an edge uses only retained square pixels with no wrap, reflection, completion, renormalization, or edge correction.
SCI-JINC-PRED-020	A supported constant field at an in-map cropped edge still returns the constant through the actual retained N_p/C_p terms.
SCI-JINC-PRED-021	$T_p^{(\kappa^2)}$ is unchanged by changing ω_i alone, receives zero from an analytic zero, and receives a positive term from either sign of nonzero κ_{ip} .
SCI-JINC-PRED-022	Missing coefficient selection by the accepted request or one exact owner-authorized versioned mode policy, an unregistered family, missing JINC permission, or an unavailable/mismatched payload prevents the affected numerical bundle without an ambient, unity, or alternate-family fallback.
SCI-JINC-PRED-023	Absence of an authorized TolTEC parameter set prevents the numerical bundle; no inherited 45 m denominator, shape values, mode-dependent $r_{\max}$, or other implementation value is substituted.
SCI-JINC-PRED-024	Permuting arrays, networks, detector rows, map groups, or container positions does not change array-to-parameter or array-to-bundle identity; a missing array creates no placeholder and does not invalidate another array.

Prediction	Falsifiable expected result
SCI-JINC-PRED-025	Ambiguous, duplicate, or aliased destination ownership fails the affected bundle before mutation and produces no partial winner.
SCI-JINC-PRED-026	Valid decompositions of the same observation into chunks replay the same scientific population, immutable joins, memberships, and equations from G_a . No numerical agreement claim is available without an exact $\mathcal{P}_{NA}$ and matching certificates; bitwise equality is not a scientific requirement.
SCI-JINC-PRED-027	An attempted cross-observation accumulator or map combination is rejected as outside base v0.1 rather than being assigned an inferred coadd rule.
SCI-JINC-PRED-028	A missing, duplicate, or ambiguous exact signal-coordinate association under the exact PTC r0.3 and AST r0.2 boundaries prevents the affected contribution and is not reclassified as ordinary outside support or repaired by another source identity.
SCI-JINC-PRED-029	Failure to form any one of the five required whole-product roles suppresses the bundle, whereas a locally invalid pixel does not suppress an otherwise complete role.
SCI-JINC-PRED-030	The base bundle contains exactly five numerical map-plane roles and no response, covariance, formal-weight, physical-exposure, standalone availability/support, diagnostic, generalized-provenance, or placeholder product. Its compact G_a information state is present but is not counted or published as a sixth product role.
SCI-JINC-PRED-031	With dimensionless ω_i , Q_p/C_p^2 remains dimensionless and is not reported as signal variance or used to create a formal-weight product.
SCI-JINC-PRED-032	The JINC route consumes no SCI-MAP product and no ordinary MAP admission, validity, exposure, support, response, covariance, or coaddition decision is substituted for a JINC-owned decision. A pass from JINC upstream-occurrence admission (Registry identity <code>SCI-JINC:jinc_map_contribution@1</code>) means only that the occurrence may be considered.
SCI-JINC-PRED-033	A changed byte, identity, version, compatibility, or supersession fact for the exact PTC r0.3 boundary, AST r0.2 boundary, admission profile, SCI-VAL binding, source-register successor, or profile-registry successor makes the affected route unavailable; no nearby source or name match is accepted. No ambient “current Registry” state is accepted; any changed bound object requires a versioned SCI-JINC successor. Matching SCI-VAL process bindings creates no new scientific content, coefficient family, parameter set, or adequacy profile.
SCI-JINC-PRED-034	Passing JINC upstream-occurrence admission (Registry identity <code>SCI-JINC:jinc_map_contribution@1</code>) leaves rounded-center admission, sample-pixel membership, cancellation support, numerical adequacy, numerical contribution, and bundle publication to their separate exact gates.

Prediction	Falsifiable expected result
SCI-JINC-PRED-035	A target WCS whose pixel matrix is anisotropic or skewed fails Equation (10) and cannot enter base v0.1 through a scalar pixel size. A compatible matrix preserves its exact axis ordering, signs, rotation, and handedness in phase and cache interpretation.
SCI-JINC-PRED-036	Replaying the immutable references and bounded state in G_a reproduces the exact input population, PTC/AST joins, admission, coefficient and parameter identities, target metric, center/phase/bin/cache decisions, square/edge membership, destination, lifecycle, required-role states, failure, and completion without a dense per-contribution archive.

H.1 Prediction-to-requirement trace

Predictions	Governing requirements
001–003	016–020, 026
004–005	024–025
006–008	008–009, 015, 017, 026
009–012	019–023
013	003, 006–007, 015, 019, 026, 029
014–017	007, 009–012
018–020	013–014, 016–020
021	026–027
022–023	004–007, 041
024–027	030–034, 044
028	002, 035–037, 043
029–030	028–029, 038–040, 044
031	024–025, 038
032	001, 035–038
033	002, 035, 037, 043
034	013, 019–023, 035–036
035	007, 010–014
036	031, 033, 040, 044

References

- [1] SCI-JINC scientific owner and Stage A investigators, *Signed-Coefficient JINC Observation Map-maker Scope Brief*, v0.1, 2026-08-28, exact object admitted by the SCI-JINC author packet.
- [2] Independent scientific-contract audit, *Independent Core: JINC Deposition, Normalization, Response, and Product Contract*, frozen 2026-08-03, used only under the SCI-JINC supersession cover.
- [3] F. Peter Schloerb, *Spectral Line Data Reduction at the Large Millimeter Telescope*, version 1.1, 2019-07-23, method excerpt from source pages 15–19, used only under the SCI-JINC LMT method reference cover.